

Weak Near-Threshold Interactions in $D^*D\bar$ Scattering from Lattice QCD

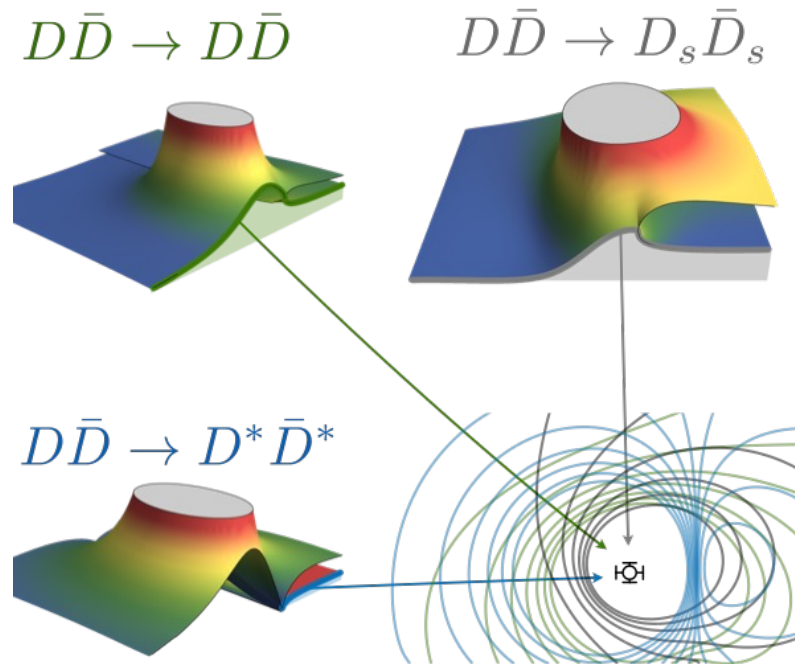


Figure representing Lattice QCD calculations of the spectrum of charm quarks – the subject of intense experimental scrutiny

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Science and Findings

- Spectrum of charmonium quark spectrum of Quantum Chromodynamics - subject of intense experimental scrutiny
- First-principles lattice QCD study of coupled $\eta_c^*\eta_c$ and $D^*D\bar$ scattering
- No evidence found for exotic charmonium resonance states near $D^*D\bar$ thresholds in contradiction to experimental determinations

Method

- Calculations used HPC systems at OLCF and JLab
- Utilizes finite-volume coupled-channel scattering amplitude formalism
- Correlation function codes developed under SciDAC

Impact

- Clean first-principles QCD baseline suggesting several experimental near-threshold “exotic” candidates may arise from production effects rather than genuine strong $D^*D\bar$ dynamics
- Challenges molecular/tetraquark interpretations
- Supports experimental programs (LHCb, Belle II, BESIII)
- Advances lattice QCD as precision tool
- Enables HPC-driven nuclear theory insights